

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

MTH 414 — Past Questions & Materials
by Renance

Course: MTH 414
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 1
Levels: 400

This study pack was compiled and restructured by Renance for MTH 414 students at Federal University of Petroleum Resources, Effurun. It brings together 1 past paper(s)/material(s) into one clean, readable document so you can revise without chasing scattered PDFs. Practise these same questions interactively on Renance with instant answers, AI explanations, and full CBT mock exams — over 235,000+ questions across JAMB, Post-UTME, WAEC, NECO and university courses.

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Paper 1 of 1 — MTH 414 Mathematical Method 11 First Semester 2020 2021

Level 400 • First Semester • PastQuestion

FIRST SEM

EMESTER EXAMINATIONS FOR 2020/2021 SESSION

MTH414 (Mathematical Methods

INSTRUCTION. ris IT) TIME: 2 1/4 Hours

swer any FOUR questions.

Question One

a. | "el) | |

State the three special cases of the Euler-Lagrange equation for finding the extremum or stationary values of the integral (? Mks)

x²

1= | F(y, y',x)ax

xy

b. Show that the shortest curve joining two points on the xy-plane is a straight line. (9 Mks)

Question Two

State and prove the first derivative property of Fourier cosine and sine transforms.

Hence or otherwise, obtain F.(f '(x)) and F,(f (x)). (18 Mks)

Question Three |

a. What is calculus of variation? (4 Mks)

b. Show that the functional

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Such that x(0) = 0, x (=) = -1, y(0)=0, y (=) = 1 is extremum for

x = -Sint and y=Sint. (14 Mks)

Question Four

a. What do we mean by Isoperimetric problem? (4 Mks)

b. Find the extrema if

™ | bo — nw*y?|dx

where f, 77 and w are given constants

(i) if y(1) = 1, but y(0) 1s not prescribed

Gi) ifneither y(0) nor y(1) are prescribed

(7 Mks)

(7 Mks)

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d Find the Fourier sine and cosine transform of the function (8 Mks)

| (By? $0 < x < \infty$)

} Pa to $x > a$

b. Find the Fourier cosine and sine transforms of the exponential function

$f(x) = e^{-ax}$, $a > 0$. (10 Mks)

Question Six

a. Find the Fourier integral representation of the function $f(x) = 1$ if $|x| < 1$.

(8 Mks)

b. A curve C joining points (x, y) and $(x, 2)$ is revolved about the x -axis.

Find the shape of the curve so that the surface generated is minimum. (10 Mks)